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Game Theory

Problem Set 1

Deadline Nov 2, 2026, 09:00, OLAT upload*

Problem 1: Four piles of matches (a version of Nim) . . . . . .

Four piles of matches are on a table. Two piles contain one match each, and another two piles contain two matches each. Two players alternately remove matches from the table. In each move, the player whose turn it is to act at that move may remove *up to two* matches from one and only one pile, and must remove at least one match. The player who removes the last match *loses* the game (contrary to the example in the lecture).

1. Describe the situation as an extensive-form game.
   Hint: It could be helpful to write the remaining number of matches in each pile on every node.
2. Does a strategy exist for one of the players that guarantees the win of that player? If yes, describe the strategy and highlight it in the game tree. If no, provide an argument why such a strategy cannot exist.

Problem 2: Game of Chomp . . . . . .

In the game *Chomp*, players take turns of biting off of a chocolate bar with $m \times n$ pieces. Each turn a player has to pick exactly one piece of the remaining bar that he wants to eat. All remaining pieces above and on the left of this piece will be removed as well. The player eating the last, poisoned piece loses the game. Consider the figure below for an example of legal moves on a 4×8 chocolate bar.

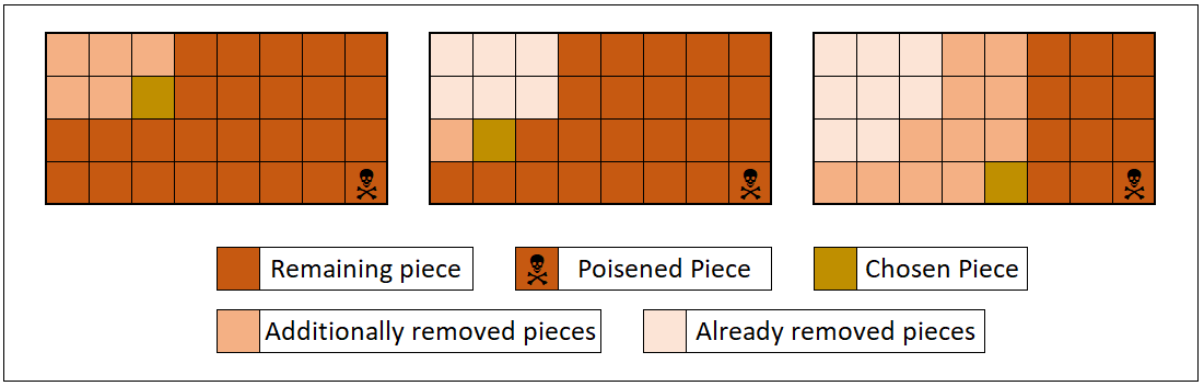


Figure 1: Legal moves in *Chomp*

*Naming scheme: *gt_ws26_ps1_matrikel1_matrikel2_matrikel3.pdf* (use your 7-digit matriculation number, **not** your "s-number" or library ID).

1. Consider a 2×4 (2 rows, 4 columns) game of *Chomp*. Does there exist a winning strategy for the first mover? If yes, define it. Otherwise provide an argument why such a strategy cannot exist.
2. Provide a formal argument why player 2 never has a winning strategy for any game size $m \times n$ except for $m = n = 1$.
   Hint: By contradiction.